

Capture the Flag 5: Make the System Interactive

Goal: Transform your Student Directory into an interactive application where administrators can favorite students, edit information, and remove records using Flutter interactions and state management.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before adding interactions.

Create:

`feature/student-interactions`

Screenshot:

- Terminal showing your current branch

```
oraiz@LAPTOP-SOMNC045 MINGW64 ~/Desktop/ulayjowonbkup/New folder/Gayl/FLUTTER/IT314A8_ORAI2 (feature/student-list)
$ git checkout -b feature/student-interactions
Switched to a new branch 'feature/student-interactions'
```

🚩 Flag 1 — Wake Up the Buttons

Add at least two buttons inside each Student Card.

Example actions:

- Favorite
- Edit

Objective: Make your Student Card respond to button presses using `onPressed`.

Screenshot:

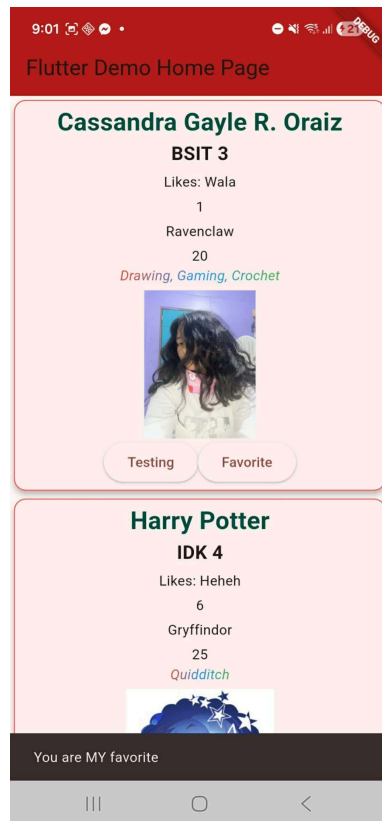
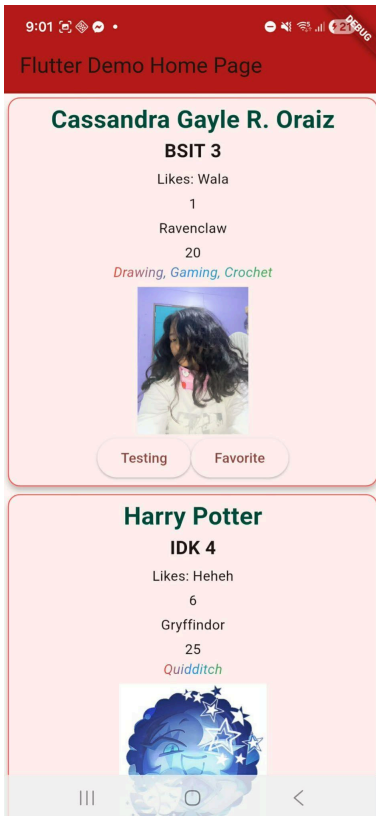
- Student Card with buttons
- `onPressed` inside your code
- Button reacting or terminal message

```
        SizedBox(height: 5),
        Image.asset(student.images, height: 150, width: 150),
        ElevatedButton(
          onPressed: () {
            ScaffoldMessenger.of(
              context,
            ).showSnackBar(SnackBar(content: Text('Button was pressed!')));
          },
          child: Text('Testing'),
        ), // ElevatedButton
      ],
    ), // Column
  ); // Card
```

```

Row(
  mainAxisAlignment: .center,
  children: [
    ElevatedButton(
      onPressed: () {
        ScaffoldMessenger.of(context).showSnackBar(
          SnackBar(content: Text('Button was pressed!')),
        );
      },
      child: Text('Testing'),
    ), // ElevatedButton
    ElevatedButton(
      onPressed: () {
        ScaffoldMessenger.of(context).showSnackBar(
          SnackBar(content: Text('You are MY favorite!')),
        );
      },
      child: Text('Favorite'),
    ), // ElevatedButton
  ],
), // Row

```



🚩 Flag 2 — Touch Anywhere

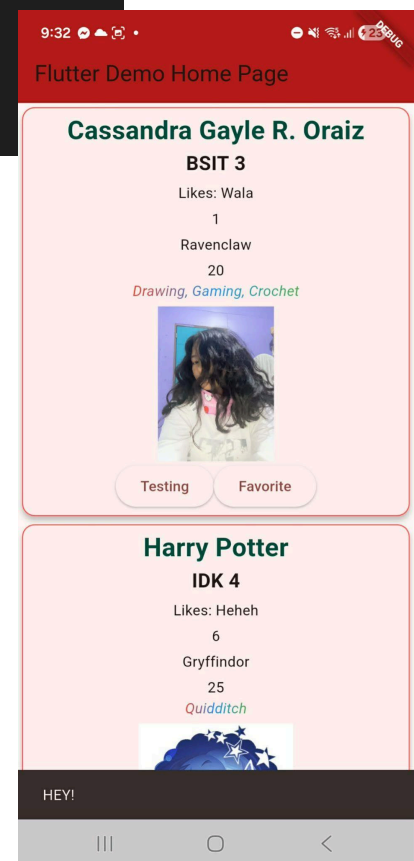
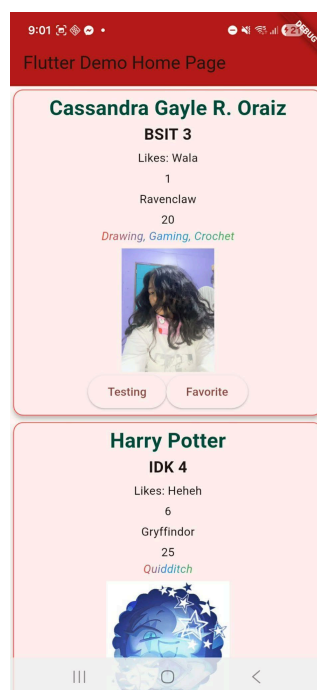
Objective: Make the entire Student Card tappable using `onTap`.

Screenshot:

- `onTap` in your code
- Student Card being tapped

```
class _MyHomePageState extends State<MyHomePage> {  
  Widget weBuildThings(Student student) {  
    return GestureDetector(  
      onTap: () {  
        ScaffoldMessenger.of(  
          context,  
        ).showSnackBar(SnackBar(content: Text('HEY!')));  
      },  
      child: Card(  
        elevation: 4,  
        shape: RoundedRectangleBorder(  
          side: BorderSide(color: Colors.redAccent, width: 1),  
          borderRadius: BorderRadius.circular(12),  
        ), // RoundedRectangleBorder  
        child: Column(  
          mainAxisAlignment: MainAxisAlignment.center,  
          children: [  
            SizedBox(height: 5),  
            Text(  

```



🚩 Flag 3 — Discover `setState`

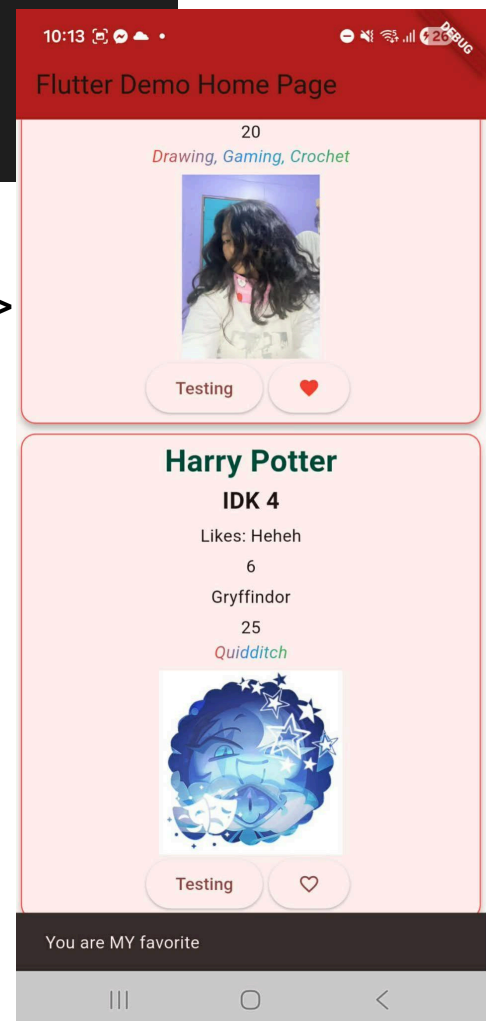
Objective: Update the screen immediately using `setState`.

Screenshot:

- `setState` in your code
- Before and after interaction

```
ElevatedButton(  
  onPressed: () {  
    setState(() {  
      student.fav = !student.fav;  
    });  
    ScaffoldMessenger.of(context).showSnackBar(  
      SnackBar(content: Text('You are MY favorite')),  
    );  
  },  
  child: student.fav  
    ? Icon(Icons.favorite, color: Colors.red)  
    : Icon(Icons.favorite_border),  
)  
  ,  
  SizedBox(width: 5),  
)
```

//Set state turns the heart button from outline to red =>

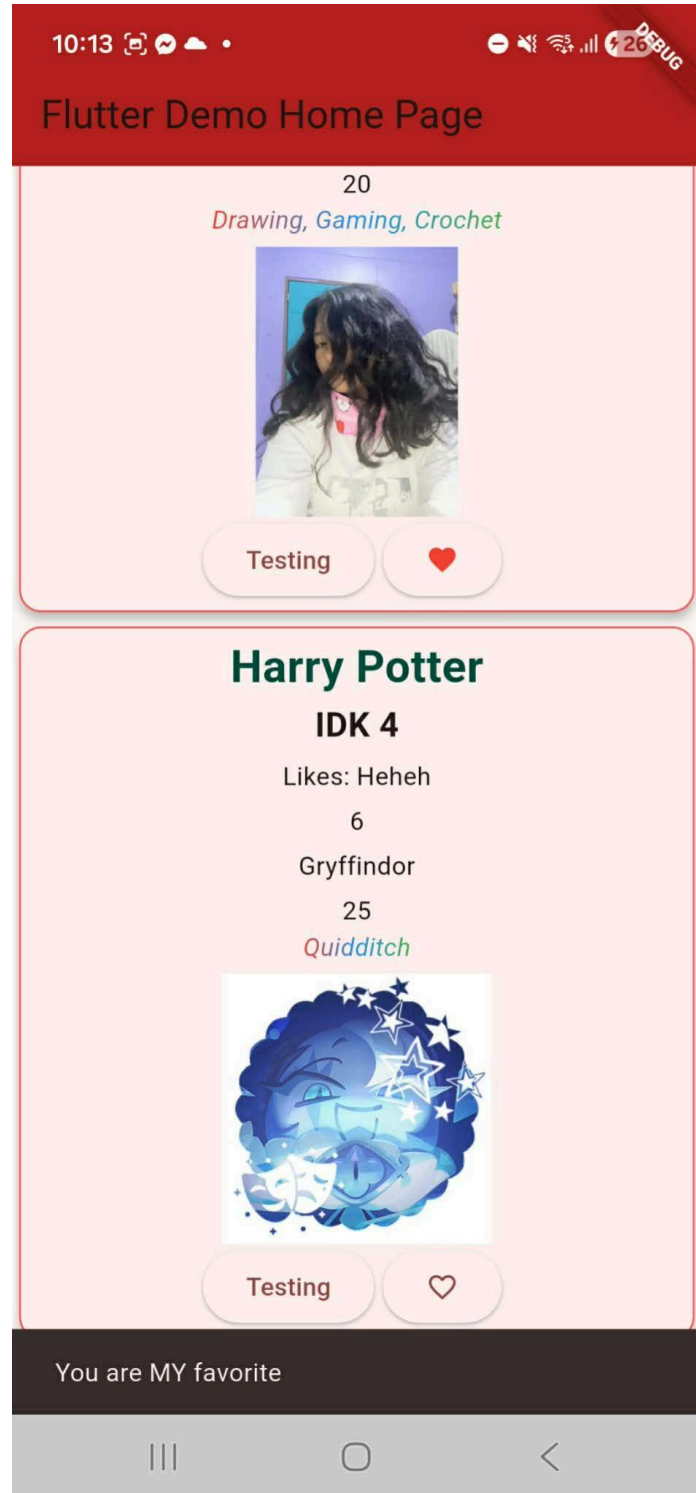


🚩 Flag 4 — Favorite Students

Objective: Let administrators mark students as favorites.

Screenshot:

- Favorited student
- Non-favorited student
- Favorite icon changing



🚩 Flag 5 — Remove a Student

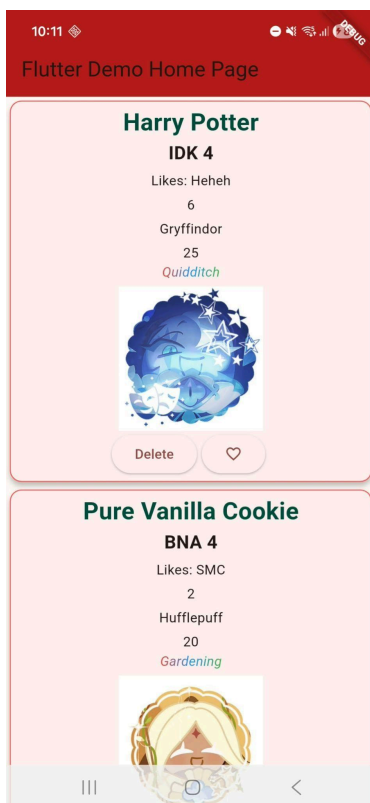
Objective: Delete a student from the directory.

Screenshot:

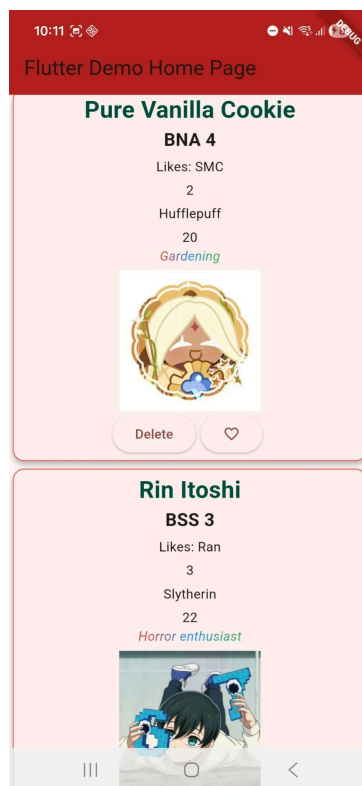
- Student before deletion
- Student after deletion

```
Row(  
  mainAxisAlignment: .center,  
  children: [  
    ElevatedButton(  
      onPressed: () {  
        setState(() {  
          students.remove(student);  
        });  
        ScaffoldMessenger.of(context).showSnackBar(  
          SnackBar(content: Text('Button was pressed!')),  
        );  
      },  
      child: Text('Delete'),  
    ), // ElevatedButton  
  ],  
  SizedBox(width: 5),  
)
```

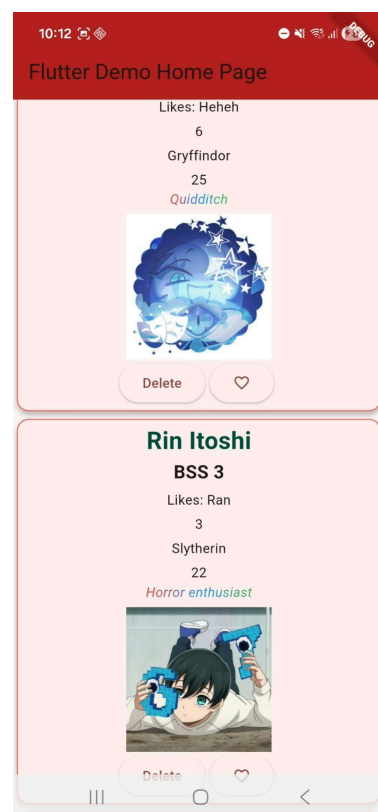
Top View:



Bottom View:



PV deleted: ('couldn't catch del msg')

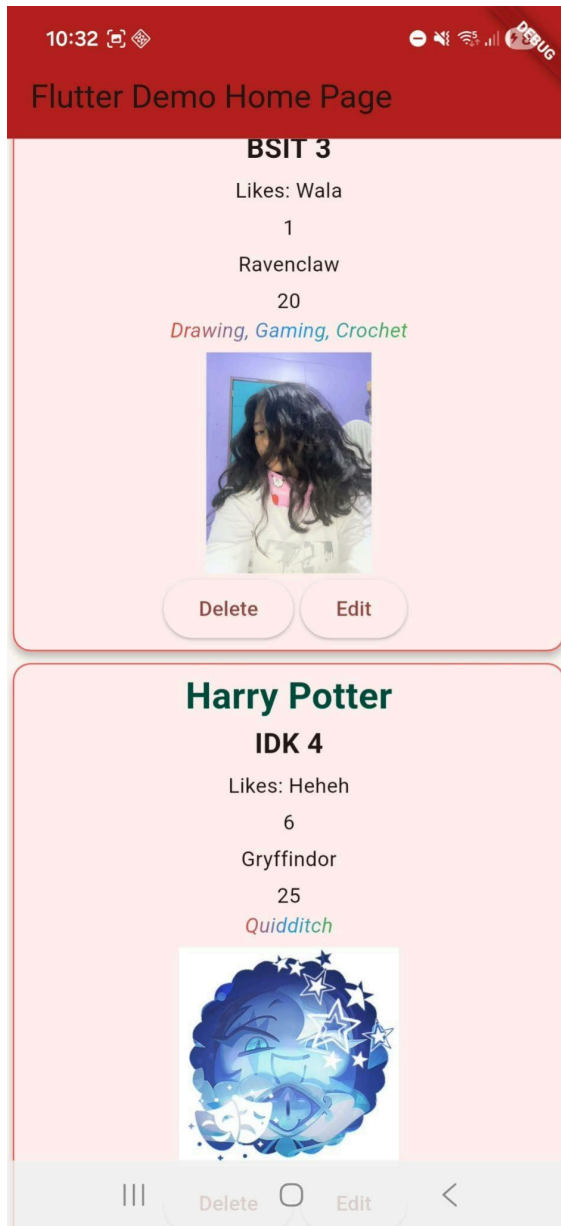


🚩 Flag 6 — Edit Trigger

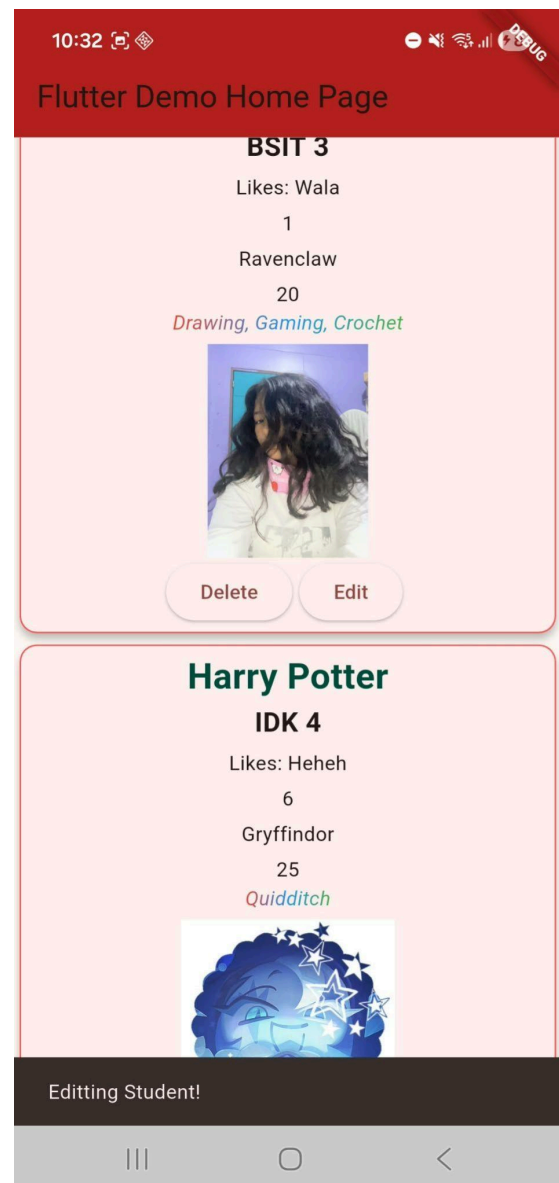
Objective: Prepare your app for editing student information.

Screenshot:

- Edit button
- Opened dialog, snackbar, or bottom sheet



```
Row(  
  mainAxisAlignment: .center,  
  children: [  
    ElevatedButton(  
      onPressed: () {  
        setState(() {  
          students.remove(student);  
        });  
        ScaffoldMessenger.of(context).showSnackBar(  
          SnackBar(content: Text('Button was pressed!')),  
        );  
      },  
      child: Text('Delete'),  
    ), // ElevatedButton  
  ],  
  SizedBox(width: 5),  
  ElevatedButton(  
    onPressed: () {  
      setState(() {});  
      ScaffoldMessenger.of(context).showSnackBar(  
        SnackBar(content: Text('Editing Student!')),  
      );  
    },  
    child: Text('Edit'),  
  ), // ElevatedButton  
),
```

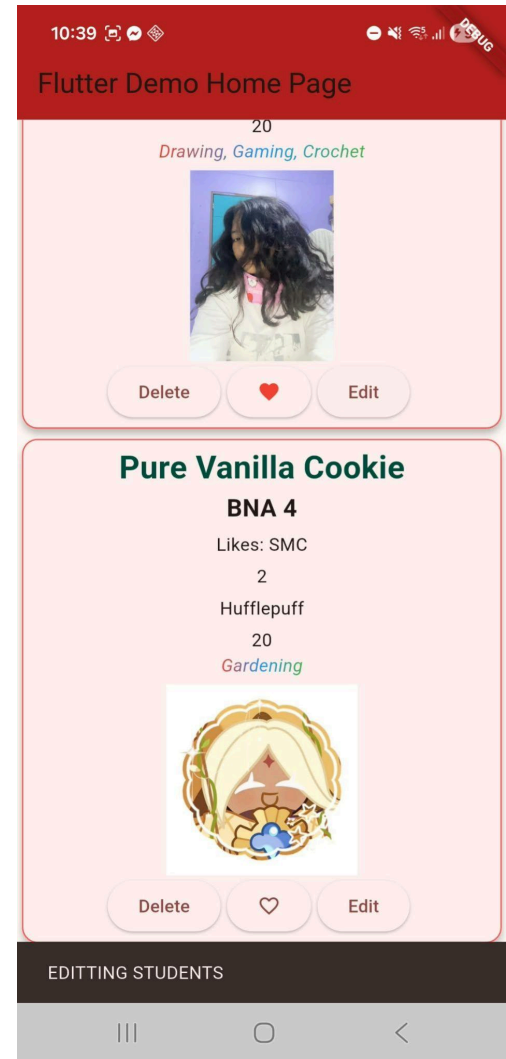
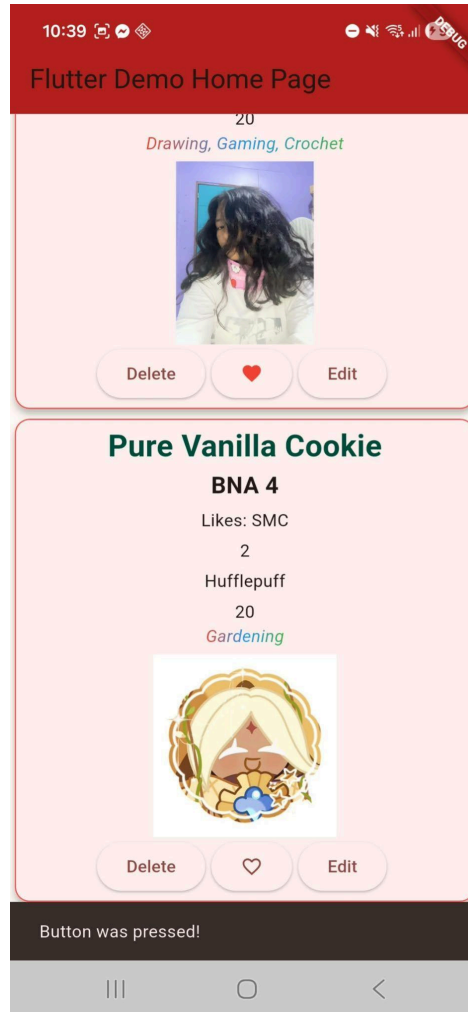
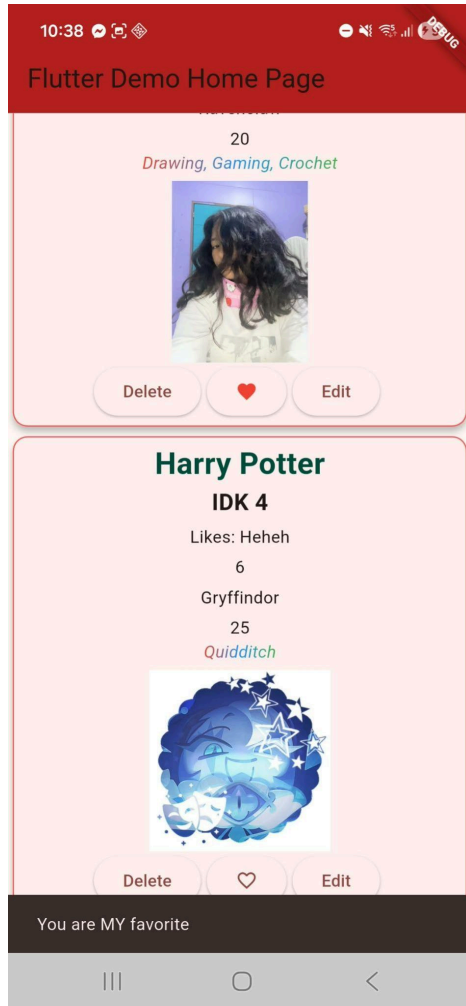


🚩 Flag 7 — Multiple Actions, Independent State

Objective: Make every Student Card manage interactions correctly.

Screenshot:

- Multiple interactions working together
- Application still functioning normally



🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Interactive Student Directory.

Use a proper Git commit message.

Branch:

`feature/student-interactions`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
oraiz@LAPTOP-SOMNC04S MINGW64 ~/Desktop/ulayjowonbkup/New folder/Gayl/FLUTTER/IT314AB_ORAIZ (feature/student-interactions)
$ git push origin feature/student-interactions
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 502 bytes | 502.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/Cassandra-Oraiz/IT314AB_ORAIZ.git
   8d5638b..2a4f2fe feature/student-interactions -> feature/student-interactions
```